# Discrete-Time Fourier Transform Properties

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# Key Concepts

1) The properties of the discrete-time Fourier transform (DTFT) help us develop insight into the nature of frequency-domain representations for discrete-time signals.

2) Recall (see *The Four Fourier Representations*) that the DTFT is $2\pi$ periodic: $X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$ so we normally consider one period or $-\pi < \omega < \pi$.

3) The DTFT is a linear operation so superposition holds.

$$ax[n] + by[n] \xleftrightarrow{DTFT} aX(e^{j\omega}) + bY(e^{j\omega})$$

The Fourier transform of a weighted sum of signals is the weighted sum of the individual Fourier transforms.

4) Symmetry for real valued $x[n]$:

   a) Conjugate symmetry: $X(e^{-j\omega}) = X^*(e^{j\omega})$.

   b) Real$\{X(e^{j\omega})\}$ = Real$\{X(e^{-j\omega})\}$. The real part has even symmetry.

   c) Imag$\{X(e^{j\omega})\} = -$ Imag$\{X(e^{-j\omega})\}$. The imaginary part has odd symmetry.

   d) $|X(e^{j\omega})| = |X(e^{-j\omega|})|$. The magnitude has even symmetry.

e) $\arg\{X(e^{j\omega})\} = -\arg\{X(e^{-j\omega})\}$. The phase has odd symmetry.

5) Time shift:

$$x[n - n_o] \xleftrightarrow{DTFT} e^{-j\omega n_o} X(e^{j\omega})$$

A shift in time introduces a phase shift that is a linear function of frequency.

6) Frequency shift:

$$e^{j\gamma n} x[n] \xleftrightarrow{DTFT} X(e^{j(\omega-\gamma)})$$

Multiplication by complex sinusoid in time introduces a frequency shift.

7) Parseval's Theorem:

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

The energy in the time- and frequency-domain representations is identical.

8) Convolution-multiplication

$$x[n] * y[n] = \sum_{k=-\infty}^{\infty} x[k]y[n-k] \xleftrightarrow{DTFT} X(e^{j\omega})Y(e^{j\omega})$$

Convolution of time-domain signals corresponds to multiplication of the corresponding DTFTs.

9) Multiplication-convolution

$$x[n]y[n] \overset{DTFT}{\longleftrightarrow} = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\gamma})Y(e^{j(\omega-\gamma)})d\gamma$$

Multiplication of time-domain signals corresponds to convolution of the corresponding DTFTs.

# Video Screenshots

DTFT: analysis of discrete-time signals and systems

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n} \quad \overset{\text{DTFT}}{\longleftrightarrow} \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

DTFT Properties - give insight and understanding
- very similar to Fourier transform properties

Linearity

$$a\,x[n] + b\,y[n] \overset{\text{DTFT}}{\longleftrightarrow} a\,X(e^{j\omega}) + b\,Y(e^{j\omega})$$

Symmetry: Suppose $x[n]$ is real

$$X^*(e^{j\omega}) = \left(\sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}\right)^* = \sum_{n=-\infty}^{\infty} x^*[n]\, e^{j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} x[n]\, e^{-j(-\omega)n} = X(e^{-j\omega})$$

$\text{Real}\{X(e^{j\omega})\}$ is even

$\text{Imag}\{X(e^{j\omega})\}$ is odd

$|X(e^{j\omega})|$ is even

$\arg\{X(e^{j\omega})\}$ is odd

# Time Shift.

$$y[n] = x[n-n_0] \overset{DTFT}{\longleftrightarrow} Y(e^{j\omega}) = e^{-j\omega n_0} X(e^{j\omega})$$

shift time by $n_0$ $\overset{DTFT}{\longleftrightarrow}$ shift phase by $-\omega n_0$ (linear)

Example: Find DTFT of $z[n] = \left(\frac{1}{4}\right)^n u[n-3]$.

Note: $z[n] = \left(\frac{1}{4}\right)^3 \left(\frac{1}{4}\right)^{n-3} u[n-3]$ and $\left(\frac{1}{4}\right)^n u[n] \overset{DTFT}{\longleftrightarrow} \frac{1}{1-1/4 e^{-j\omega}}$

So $Z(e^{j\omega}) = \left(\frac{1}{4}\right)^3 e^{-j3\omega} \left(\frac{1}{1-1/4 e^{-j\omega}}\right)$

# Frequency shift.

Let $y[n] = e^{j\gamma n} x[n]$

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} e^{j\gamma n} x[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x[n] e^{-j(\omega-\gamma)n} = X(e^{j(\omega-\gamma)})$$

$$e^{j\gamma n} x[n] \overset{DTFT}{\longleftrightarrow} X(e^{j(\omega-\gamma)})$$

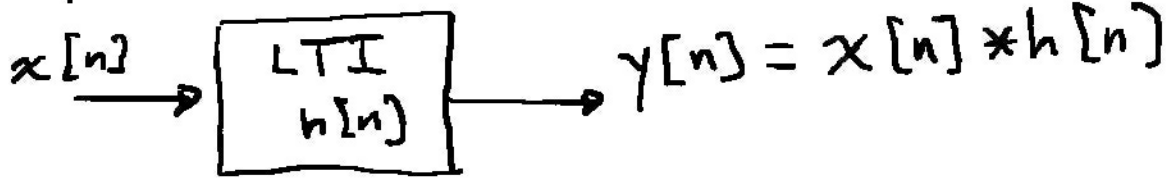
Example: $z[n] = (-1)^n (1/4)^n u[n] = e^{j\pi n} \left(\frac{1}{4}\right)^n u[n]$

$$Z(e^{j\omega}) = \frac{1}{1 - 1/4 e^{-j(\omega-\pi)}} = \frac{1}{1 + 1/4 e^{-j\omega}}$$

# Convolution - Multiplication

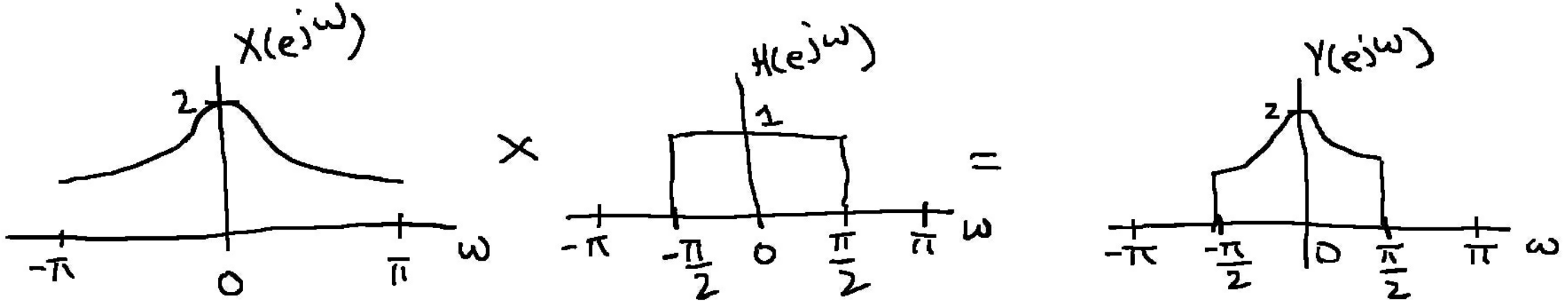
$$x[n] * y[n] = \sum_{k=-\infty}^{\infty} x[k]\, y[n-k] \quad \xleftrightarrow{\text{DTFT}} \quad X(e^{j\omega})\, Y(e^{j\omega})$$

Example:




$$h[n] = \frac{\sin\left(\frac{\pi}{2} n\right)}{\pi n}$$

$$Y(e^{j\omega}) = X(e^{j\omega})\, H(e^{j\omega}), \qquad H(e^{j\omega}) = \begin{cases} 1 & -\pi/2 < \omega < \pi/2 \\ 0 & \text{otherwise on } (-\pi, \pi] \end{cases}$$

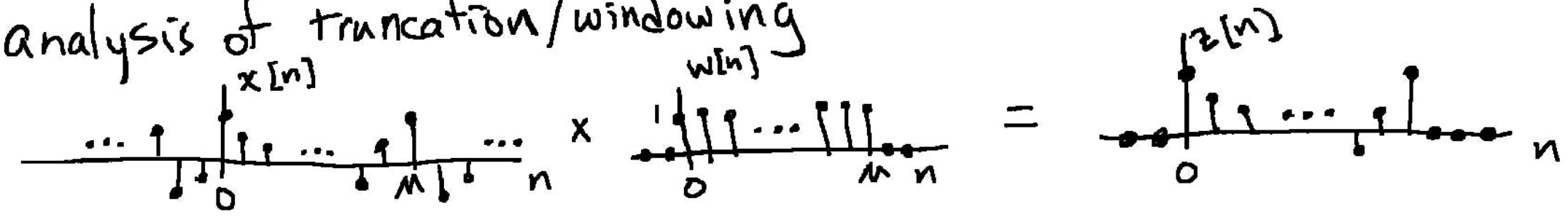



Lowpass

# Multiplication – Convolution (windowing)

$$z[n] = x[n]\,w[n] \quad \overset{\text{DTFT}}{\longleftrightarrow} \quad Z(e^{j\omega}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\gamma})\,W(e^{j(\omega-\gamma)})\,d\gamma$$

analysis of truncation/windowing




## Parseval's Theorem

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} |X(e^{j\omega})|^2\,d\omega$$

energy is conserved!

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